The Fertility Race Between Technology and Social Norms*

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Abstract

This paper studies fertility as the outcome of a tug-of-war between gender-biased technological progress and social norms around the gender division of childcare. We construct a fertility choice model that features endogenous social norm formation, shaped by the collective opinions of all cohorts. Consistent with our model, cross-country data show that fertility declines more rapidly in economies undergoing rapid structural transformation, and this relationship is more pronounced in societies with tighter or less adaptive social norms. Calibrated to South Korean data, our model indicates that in the context of rapid structural transformation, strong social pressure and the reluctance to adapt exacerbate fertility declines and reinforce rigid norms. We also demonstrate that while subsidies for female childcare offer larger short-run fertility increases, male childcare subsidies yield greater long-term gains by accelerating the transition to a more egalitarian steady state.

JEL classification: J12, J13, O11

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1. Introduction

The drastic decline in fertility rates below the replacement level poses an imminent challenge for many economies over the coming decades, threatening fiscal sustainability and economic growth (Jones 2022). This challenge is especially conspicuous in nations "getting old before getting rich" where low fertility imperils catching-up prospects. Uncovering the causes of low fertility, forecasting its path, and pinpointing effective policy solutions are thus critical questions for researchers and policymakers.

In this paper, we develop a quantitative overlapping-generations model in which fertility emerges from a tug-of-war between gender-biased technological progress and evolving social norms on the gender division of childcare responsibilities. In our model, fertility decisions follow the limited commitment framework by Doepke and Kindermann (2019) where parents negotiate childcare allocation and consumption. Extending their framework, we assume that parents determine the childcare allocation by balancing the opportunity costs of childcare against the utility losses of deviating from the prevailing social norm. For example, in a society with traditional gender expectations, assigning more childcare to men might lower costs if women start to earn higher wages, yet it could provoke social disapproval, adding utility losses. Thus, social norms create a wedge between the actual childcare arrangement and the one that would minimize opportunity costs alone, raising the shadow price of children for parents. As a result, these norms significantly influence fertility and labor supply in the model.

Different from prior studies that treat social norms as exogenously given, our key contribution is to explicitly model how social norm is endogenously formed. Inspired by the insights from the prior literature (e.g., Bisin and Verdier 2001, Brooks and Bolzendahl 2004), we hypothesize that the prevailing norm reflects a weighted average of the *opinions* expressed by the other members in the society. These opinions, in turn, are formed in a re-evaluation stage and depend on the current economic conditions as well as the childcare practices that these members adopted in the past. Therefore, the evolution of the social norm in the economy stems from two sources. On the one hand, when new cohorts enter the economy and adopt childcare practices that are different from the past, they shift future norms, reflecting *between-cohort effects* (also known as cohort replacement effects). On the other hand, older cohorts re-evaluate the current situation and weigh it against their past practices. As a result, changing economic conditions affect the opinions expressed by older cohorts, reflecting *within-cohort effects* (also known as social structural effects). Furthermore, the influence of the two effects depends on the population weight of the corresponding cohort, which in turn hinges on past fertility choices.

The model suggests that strong social pressure and older cohorts' reluctance to re-evaluate impose rigid constraints on young parents' choices, fostering cultural inertia. If technological advancements necessitate a shift in childcare allocation, this inertia prevents adjustment, increasing the burden on one side of the parents and making childbearing costlier for them. Given that fertility requires both parents' consent, this added cost can sharply reduce fertility rates, a pattern consistent with our empirical evidence. Our model also predicts a non-monotonic fertility trajectory: an initial drop followed by a recovery as norms adapt to economic realities, in line with the arguments in Feyrer et al. (2008) and Doepke et al. (2023). More importantly, it also offers quantitative predictions about the timing of this fertility recovery and the pace of norm evolution, driven by both within- and between-cohort effects.

Consistent with the model's predictions, we document that countries undergoing rapid structural change–one form of gender-specific technological changes as shown in Ngai and Petrongolo (2017)–experience a more pronounced decline in fertility rates. This relationship is particularly pronounced in societies with tighter or less adaptive social norms. Specifically, our cross-country regression analysis reveals that a 1-percentage-point increase in the pace of service sector expansion per year is associated with an annual fertility decline of 0.1 children per woman. This correlation is 50% stronger in societies exhibiting above-median cultural rigidity, as measured by the tightness-looseness index from Uz (2015). The correlation is also stronger in societies with less adaptive social norms, as measured by the overall changes in attitudes over time. The pattern persists even after accounting for GDP per capita growth and remains consistent when we substitute agricultural sector decline as a measure of structural change.

Furthermore, to evaluate the model's quantitative predictions, we calibrate the model to South Korean experience from 1976 to 2016. Gender-specific wage paths are set to align with trends in total factor productivity and gender wage gaps. Preference parameters are adjusted to match fertility trends, while the substitutability of childcare between genders reflects the initial allocation. The strength of social pressure is calibrated to fit gender disparities in childcare time from the Korean Time Use Survey (KTUS), yielding estimates consistent with prior research (e.g., Park 2021). The weight that old cohorts put on their past childcare practices when they re-evaluate the current condition is calibrated to match the share of within-cohort changes in driving the overall social norm evolution, calculated using data from the Korean General Social Survey (KGSS).

The calibrated model indicates that social norms play an important role in mediating the fertility effects of gender-biased technological changes. For example, we examine the fertility responses to the changing gender wage gap in the sample period under two scenarios. In the first scenario, we fix the social norm at the initial steady state level. In the second scenario, we allow the social norm to adjust to the level of the new steady state. We find that the fertility decline in the first scenario is almost twice as large as that in the second scenario.

Using the calibrated model, we conduct five counterfactual experiments. In the first counterfactual, we adjust the pace of structural change in the economy. We find that faster structural change accelerates the fertility decline, aligning with empirical evidence. However, the transition paths of childcare and social norms remain largely unchanged, as strong social pressure limits shifts in childcare allocation during this transformation.

We then experiment with the roles of social norms. In the second counterfactual, we vary the intensity of social pressure while holding the structural change constant. Here, economies with weaker social pressure exhibit a smaller fertility drop and a faster social norm transition toward a new steady state, consistent with our empirical findings. In the third counterfactual, we vary the weight that older cohorts put on their past childcare practices when they re-evaluate the current situation. We find that a smaller weight accelerates the social norm convergence towards the new steady state and raises fertility rates along the transition path.

In the fourth counterfactual, we examine the hypothetical fertility transition that would occur if we adjust the parameters calibrated for South Korea to those of the U.S.. We find lower social pressure and resistance to change in the U.S. compared to South Korea. Using the U.S. parameters, we estimate that South Korea would have experienced a less severe fertility decline and a more rapid recovery, with accelerated adjustment of social norms.

In our last counterfactual experiment, we evaluate the impacts of gender-specific childcare subsidies. Our findings indicate that while subsidies for female childcare yield greater short-term increases in fertility, subsidies for male childcare produce significantly larger long-term fertility gains. This is because male childcare subsidies accelerate the transition to a more egalitarian long-run steady state, leading to increased policy exposure over time and higher desired fertility among females due to improved social norms. These predictions are in line with the recent empirical evidence from Lee and Lee (2025) and Kleven et al. (2025).

To summarize, this paper develops a model of fertility choice under endogenous social norm formation. We find support for the model's key predictions using cross-country data. Then, calibrated to South Korea, the model matches observed trends, delivers new predictions (non-monotonic fertility path), and highlights a novel policy mechanism through the endogenous social norm transition.

Related Literature

This paper contributes to the growing body of literature examining the effects of technological change on fertility and female labor supply. The most closely related studies in this domain are Goldin (2024), Doepke and Kindermann (2019), Myong et al. (2021), and Fogli and Veldkamp (2011).[1]

The study by Goldin (2024) illustrates that countries with "lowest-low" fertility rates experienced rapid per capita GNP growth alongside gradually evolving cultural traditions. The author proposes a theoretical framework in which generational and gender-based conflicts drive a significant decline in total fertility rates. Our research aligns with this perspective, arguing that fertility rates result from a dynamic interplay between gender-biased technological progress and societal norms. However, our work differs from Goldin (2024) in several key aspects. First, in contrast to Goldin (2024), which assumes gender-specific desired fertility levels, our model derives fertility decisions and bargaining dynamics between women and men from microeconomic principles. Second, we develop a model of social norms that incorporates both within- and between-cohort effects, allowing economic agents to adapt flexibly to changing economic conditions rather than adhering strictly to traditional norms. Third, while Goldin (2024) suggests that men prioritize inherited traditions more than women, resulting in a higher desired number of children, we assume uniform preferences across genders and attribute differences in desired fertility to negotiations over childcare responsibilities and initial technological conditions. Lastly, we establish a new empirical finding by demonstrating a strong correlation between structural economic change and fertility decline, even

[1] See Doepke et al. (2023) for an excellent summary of the recent literature.

after controlling for GNP growth, which is the primary focus of Goldin (2024).

Doepke and Kindermann (2019) find that countries with more equitable childcare practices exhibit fewer discrepancies between men and women regarding desired fertility, resulting in higher birth rates. They explain this pattern with a bargaining model with limited commitment. We build on and extend their framework by incorporating the influence of social norms and modeling the norms' endogenous evolution. This additional element allows our model to address how economies adjust to technological changes over time through social norm adaptation.

The study by Myong et al. (2021) explores fertility and marriage patterns in East Asian societies, analyzing the effects of two social norms: one related to the stigma of out-of-wedlock births and another concerning the gender division of childcare. Estimating their model with South Korean data, they conclude that the latter norm exerts a stronger influence on fertility outcomes. This finding inspires our focus on the childcare social norm, abstracting from marriage norm considerations.[2] While Myong et al. (2021) treat the childcare social norm as exogenous and estimate it empirically, we micro-found the endogenous formation of such norms. Consequently, our approach enables predictions about future fertility trends and facilitates counterfactual analyses based on structural parameters of norm formation.

Our research complements studies on the evolution and transmission of social norms, such as Bisin and Verdier (2001), Bisin and Verdier (2011), and Giuliano and Nunn (2021), and the role of culture in shaping fertility transition, such as Spolaore and Wacziarg (2022) and Beach and Hanlon (2023). One particularly related paper is by Baudin (2010) who adopts a Bisin-Verdier model to study the role of cultural transmission, in particular "traditionalist" versus "modernist", in shaping fertility transition after the Industrial Revolution. Our paper complements Baudin (2010) by focusing on the gender division of childcare responsibilities and the associated implications on fertility. We also make a methodological contribution in the modeling of social norms. Whereas the influential Bisin-Verdier framework emphasizes parents' forward-looking socialization decisions during the child-rearing period, we abstract away from the forward-looking aspect but assume that older cohorts in the economy solve an opinion expression problem each period. This assumption allows for frequent interactions across generations, instanta-

[2]Similar to Myong et al. (2021), we also abstract away from status externalities in Kim et al. (2024) or the fertility norm in De Silva and Tenreyro (2020).

neous reactions of all agents to technological shocks, and more numerical tractability.

Lastly, several influential papers have explored social learning as a key mechanism driving the increase in female labor force participation and changing marriage patterns amid technological advancements (e.g., Fernández et al. 2004, Fernández and Fogli 2009, Fogli and Veldkamp 2011, Fernández 2013, Bertrand et al. 2021). These studies highlight how individuals learn about the effects of female labor force participation on child outcomes and how this knowledge is transmitted across generations (vertical socialization) and within neighborhoods or peer groups (horizontal socialization). Complementing their approach, we investigate the role of social norms surrounding childcare responsibilities, examining their impact on household decisions and their evolution through both within-cohort and between-cohort adjustments, ultimately driven by technological changes.

The remainder of the paper is structured as follows. Section 2 outlines the theoretical model, its core mechanism, and predictions. Section 3 presents the supporting evidence. Section 4 details the calibration strategy and results using South Korean data. Section 5 conducts the primary counterfactual exercises. Finally, Section 6 offers concluding remarks.

2. Model

This section introduces an overlapping generations model where parents bargain over fertility and childcare responsibilities under the influence of social norm. Moreover, the prevailing social norm in the economy is endogenously determined by older cohorts' opinions which depend on past childcare practices.

2.1 Household Problem

We analyze an overlapping generations economy where women and men live until age J. Each cohort consists of women and men of equal mass.[3] To examine generational conflicts and fertility dynamics, we assume homogeneity among agents of the same gender within each cohort. To emphasize fertility and childcare decisions, we model individuals as consuming their labor income before and after age J_f, the period during

[3]The size of each incoming cohort may vary over time due to endogenous fertility decisions.

which individuals form couples and engage in a bargaining problem.

A couple comprises a woman and a man, indexed by gender $g \in \{♀, ♂\}$. They choose individual consumption c^g, childcare contributions l^g, and the number of children n, which is a household-shared decision. We use t to denote time. Gender-specific market wages, denoted w_t^g, are exogenous and reflect technological changes impacting labor demand differentially by gender (Ngai and Petrongolo 2017).[4]

Individual's preference over fertility and consumption is given by:

$$u^g(c^g, n) = c^g + \gamma \cdot \frac{n^{1-\rho} - 1}{1-\rho} \qquad \rho > 0 \tag{1}$$

where c^g represents personal consumption and n the number of children. The parameters γ and ρ control the weight and curvature of fertility n in the utility function, respectively.

Raising each child incurs a time cost ϕ. Thus, to support n children, the couple must satisfy the childcare provision constraint:

$$n\phi = \left(\beta \cdot (l^{♀})^{\frac{\sigma-1}{\sigma}} + (1-\beta) \cdot (l^{♂})^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1 \tag{2}$$

where $l^{♀}$ and $l^{♂}$ denote childcare time from the woman and man, respectively, β governs the relative productivity of each parent in raising children, and σ governs the substitutability of their contributions.

Following Doepke and Kindermann (2019), we assume that spouses bargain over fertility and childcare under partial commitment. The decision-making timeline is outlined below.

First Stage: Childcare Arrangements

The couple first determines a plan for childcare allocation, conditional on having n chil-

[4]Alternatively, we can model structural change via time-varying, gender-specific labor demand shifters, with equilibrium wages determined by market clearing. In the calibration, these shifters can be backed out given that we observe equilibrium wages and generate hours worked from the model. However, as this paper focuses on household-side responses to technological changes and the role of endogenous social norms, we adopt the simpler exogenous wage approach. We also abstract away from the causality in the reverse direct, i.e., declining fertility and rising female labor supply stimulates structural transformation, a possibility discussed by Kuhn et al. (2024) in line with the Rybczynski theorem.

dren, and commits to this decision post-childbirth.[5] For all n, the couple solves:

$$\min_{l^{♀}, l^{♂}} \quad w_t^{♀} l^{♀} + w_t^{♂} l^{♂} + \lambda \cdot w_t^{♂} \cdot \left(\frac{l^{♀}}{l^{♂}} - \eta_t \right)^2, \tag{3}$$

subject to the childcare constraint (2). Here, $w_t^{♀} l^{♀} + w_t^{♂} l^{♂}$ represents the opportunity cost of childcare in the units of consumption goods, while $\lambda \cdot w_t^{♂} \cdot \left(\frac{l^{♀}}{l^{♂}} - \eta_t \right)^2$ captures utility losses from deviating from the prevailing social norm, η_t, which is taken as given by parents at time t.[6] The parameter λ governs the intensity of these social pressure costs. We scale the psychic cost by $w_t^{♂}$ so that what matters for the childcare allocation is the gender wage gap $w_t^{♀}/w_t^{♂}$ rather than the universal wage levels. The solutions to this cost-minimization problem, denoted $l_t^{♀}(n)$ and $l_t^{♂}(n)$, represent optimal childcare allocation under the influence of the social norm.

Second Stage: Bargaining over Fertility

Next, the couple negotiates the number of children, n_t. Only mutually agreed-upon fertility is realized, defined as:

$$n_t = \min\{n_t^{♀}, n_t^{♂}\}, \tag{4}$$

where n_t^g is the fertility level that maximizes the ex-post utility of gender $g \in \{♀, ♂\}$ in the third stage. In other words, if we denote the solution to the consumption bargaining problem as $c_t^g(n)$ for $g \in \{♀, ♂\}$, the ex ante desired fertility level for each gender is given by

$$n_t^g = \arg\max_n \quad u^g(c_t^g(n), n). \tag{5}$$

Combining Equations (4) and (5), it is evident that when men and women exhibit differing desired fertility levels, the lower level predominates. Consequently, factors that exacerbate the disparity in desired fertility between genders, such as social norms that impede equitable childcare responsibilities, lead to reduced fertility in the equilibrium.

Third Stage: Bargaining over Consumption

[5] Doepke and Kindermann (2019) justifies this commitment by noting that childcare decisions involve significant switching costs and advance planning (e.g., securing daycare slots before birth) and interact with persistent choices like residential location, which affects childcare availability.

[6] An equivalent way to formulate the psychic cost is to put it in the individuals' preferences instead of the childcare allocation problem. In that case, Equation (1) becomes $u^g(c^g, n) = c^g + \gamma \cdot \frac{n^{1-\rho}-1}{1-\rho} + \lambda \cdot w_t^{♂} \cdot (\frac{\tilde{l}^{♀}(n)}{\tilde{l}^{♂}(n)} - \eta_t)^2$ where $\tilde{l}^g(n)$ is the solution to the opportunity cost minimization problem alone. We thank Ananth Seshadri for pointing this out.

Post-childbirth, the couple implements the agreed childcare arrangement $l_t^g(n)$ and bargains over consumption $c^g(n)$. If the couple ends up having n children, each gender's outside option in the non-cooperative case is:

$$\overline{u}^g(n) = w_t^g(1 - l_t^g(n)) + \gamma \cdot \frac{n^{1-\rho} - 1}{1 - \rho}, \qquad \rho > 0, \tag{6}$$

reflecting consumption of residual labor income after childcare. Following Doepke and Kindermann (2019), the non-cooperative state is modeled as a continuing relationship along the lines of the separate-spheres bargaining model of Lundberg and Pollak (1993). That is, the couple is still together and both partners still derive utility from the child, but bargaining regarding the allocation of consumption breaks down, the division of child care duties follows the ex-ante commitment $l_t^g(n)$, and the couple no longer benefits from returns to scale in joint consumption.

If they cooperate, the Nash bargaining problem is:

$$\max_{c^{♀}, c^{♂}} \quad \left(u^{♀}(c^{♀}, n) - \overline{u}^{♀}(n)\right)^{1/2} \cdot \left(u^{♂}(c^{♂}, n) - \overline{u}^{♂}(n)\right)^{1/2}, \tag{7}$$

subject to the budget constraint:

$$c^{♀} + c^{♂} = (1 + \alpha) \cdot [w_t^{♀}(1 - l_t^{♀}(n)) + w_t^{♂}(1 - l_t^{♂}(n))], \tag{8}$$

where α captures economies of scale from cooperation. Following Doepke and Kindermann (2019), we assign each spouse equal bargaining power (one-half).

2.2 Endogenous Social Norm

A novel feature of this model is that the prevailing social norm is endogenously determined. Unlike the exogenous social norm assumption used in the literature, we posit that the social norm at time t, denoted η_t, emerges as a weighted average of the *opinions* expressed by the other members of the society, which in turn depends on the childcare practices in the past. This dynamic process ties current norms to historical household decisions, capturing both social pressure and reevaluation by older cohorts.

Specifically, the prevailing social norm at the time t is defined as:

$$\eta_t = \sum_{j=1}^{J-J_f} \phi_{J_f+j,t} \cdot \tilde{\eta}_{J_f+j}, \qquad \sum_{j=1}^{J-J_f} \phi_{J_f+j,t} = 1, \tag{9}$$

where $\tilde{\eta}_{J_f+j}$ represents the opinions of childcare allocation expressed by the cohort at age $J_f + j$ at time t, and $\phi_{J_f+j,t}$ is the weight assigned to that cohort's influence.[7] This weight reflects the share of households aged $J_f + j$ at time t among the subpopulation older than J_f, calculated as:

$$\phi_{j,t} = \frac{\pi_{j,t}}{\sum_{k=J_f+1}^{J} \pi_{k,t}}, \tag{10}$$

where $\pi_{j,t}$ denotes the population share of the cohort aged j at time t. These weights ensure that the influence of past decisions is proportional to the relative size of each cohort.

Motivated by the "imperfect empathy" approach by Bisin and Verdier (2001), we assume that older cohorts form opinions by solving the following minimization problem:

$$\tilde{\eta}_{J_f+j} = \arg\min_{\eta} \quad \underbrace{w_t^{♀} \cdot \eta + w_t^{♂}}_{\text{opportunity cost of childcare}} \quad + \psi \cdot w_t^{♂} \cdot \quad \underbrace{\left(\eta - \frac{l_{t-j}^{♀}}{l_{t-j}^{♂}}\right)^2}_{\text{distance to own practice in the past}} \tag{11}$$

where $\frac{l_{t-j}^{♀}}{l_{t-j}^{♂}}$ measures the childcare practice adopted by these agents j periods ago. Like the parents' problem in (3), the psychic cost is scaled by $w_t^{♂}$ so that what matters for the opinion expression problem is the gender wage gap $w_t^{♀}/w_t^{♂}$. The solution to this problem reflects the *desired gender ratio of childcare* that balances the economic costs and utility losses by cohort $J_f + j$.

This minimization problem reflects changes in social norms within cohorts, where older generations re-evaluate the situation given the prevailing market wage and choose the opinion they express to the current parents. Parameter ψ governs the utility costs of deviating from their own past choice, i.e., stubbornness. If they are perfectly altruistic towards the current parents, they would simply discard their own past decisions and

[7]By assuming away heterogeneity with cohorts, this paper abstracts away from the horizontal socialization discussed in Bisin and Verdier (2011). Recently, Albrecht et al. (2024) and Lee and Lee (2025) have explored this mechanism in the setting of parental leave taking.

minimize the opportunity costs of childcare, i.e., $\psi = 0$. However, if ψ is high, they will express opinions that are more in line with how they brought up their own children j periods ago.

2.3 Population Dynamics

The demographic structure of this economy, denoted $\{\pi_{j,t}\}_{j=1}^{J}$, evolves endogenously according to a law of motion driven by the fertility rate n_t.

Let $\boldsymbol{\pi_t} = (\pi_{1,t}, \ldots, \pi_{J,t})^T$ represent the population distribution across age groups at time t, where $\pi_{j,t}$ is the share of individuals aged j. The population dynamics are governed by:

$$\boldsymbol{\pi_{t+1}} = \frac{\boldsymbol{\Pi_t} \cdot \boldsymbol{\pi_t}}{\|\boldsymbol{\Pi_t} \cdot \boldsymbol{\pi_t}\|_{L^2}}, \tag{12}$$

where $\boldsymbol{\Pi_t}$ is a $J \times J$ demographic transition matrix.

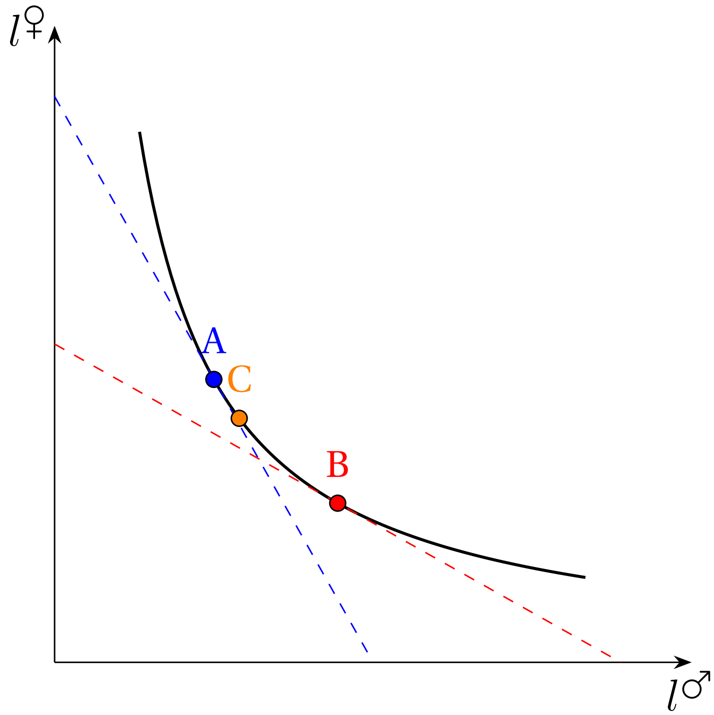
The matrix $\boldsymbol{\Pi_t}$ has two key features: (1) elements in the j-th row and $(j+1)$-th column equal 1, reflecting aging from age j to $j+1$, and (2) the element in the first row and J_f-th column equals $n_t/2.1$, representing births from couples at age J_f, normalized by the replacement fertility rate (approximately 2.1 children per couple). The L^2-norm ensures the population shares sum to 1, maintaining a normalized distribution over time.

2.4 Mechanisms

The central prediction of this model is that fertility emerges from a dynamic tension between gender-biased technological progress and evolving social norms. Technological change, by increasing women's relative wages, shifts the optimal childcare division toward greater gender equality for young couples making fertility decisions—absent social norm pressures (see Equation (3)). However, when social norms enforce traditional roles, they act as a form of "adjustment costs," generating a gap between the actual childcare arrangement and the one that would minimize opportunity costs alone, leading to a higher shadow price of children from the parents' perspective (Figure 1 illustrates the intuition of this mechanism). As a result, couples prefer fewer children ex ante. Conditional on the prevailing social norm, the strength of this channel depends on social norm tightness/pressure λ.

On the other hand, social norms in this economy are endogenous, shaped by changes

Figure 1: The Childcare Allocation Problem with Social Norms



Notes: This figure plots the childcare minimization problem faced by the household in the presence of social norms. The isoquant curve ABC represents combinations of male and female childcare hours that produce one unit of childcare. The slopes of budget lines reflect prevailing gender wage gap $l^{♂}/l^{♀}$. Point A is the original steady state where females shoulder a greater share of childcare responsibilities under a large gender wage gap. Gender-biased structural change alters the relative wage, shifting the cost-minimizing point to B in the long-run equilibrium. Due to pressures of social norms, however, the household chooses point C, closer to A, which is more costly than B under the new relative wage. This increases the shadow price of childcare and reduces fertility. As social norm evolves, the household choice converges to point B and hence fertility recovers. According to the model, point C on impact will be closer to point B if the economy has less social pressure (small λ) or less stubbornness (small ψ).

within- and between-cohort (see Equation (9)). Entering cohorts, influenced by improved labor market opportunities for women, may favor more equitable childcare arrangements. Older cohorts, aware of the changes in the gender wage gap, may also shift their positions and choose which opinions to express (see Equation (11)).

The evolution of the social norm thus stems from two sources. First, when new cohorts enter at age J_f and choose childcare contributions $l^{♀}$ and $l^{♂}$ that diverge from established practices, they shift future norms toward their preferences. For instance, a cohort opting for more egalitarian childcare ratios will incrementally reduce η_t in subsequent periods. This channel reflects between-cohort changes, also known as the cohort replacement effects.

Second, older cohorts choose the opinions they would like to express. This decision hinges on their past experiences, but also reflects their re-evaluation of the situation given the current conditions. This channel reflects within-cohort changes, also known as the social structural effects, with the strength governed by the stubbornness parameter ψ.

Note that the influence of these two effects depends on the population weight of the corresponding cohort, $\phi_{j,t}$, which is itself an endogenous outcome of fertility decisions made by their parents' generation. Higher past fertility increases a cohort's size, amplifying its impact on η_t. This interplay between fertility, demographic structure, childcare choices, and opinions drives the norm's gradual adaptation over time.

Even though the social norm adjusts through both within- and between-cohort effects, the adaptation speed might still be slower than the rapid changes in technology. Therefore, the model predicts a non-monotonic fertility response to technological change that boosts women's wages. Initially, fertility declines sharply as the social norm lags, reflecting entrenched gender roles. Over time, as the social norm adjusts toward equality, fertility recovers. This transition highlights the interplay between short-term path dependence and long-term adaptation.

Based on these observations, the model generates two testable predictions for economies experiencing gender-biased technological changes:

Prediction 1: Economies experiencing faster gender-biased technological changes exhibit more rapid fertility declines.

Prediction 2: The impact of gender-biased technological changes on fertility is stronger in economies with social norms that are either tighter (larger λ) or less adaptive (larger ψ).

The advantage of explicitly modeling the social norm is that we can test these predictions, quantify the model's channels, and make numerical predictions. Correspondingly, we provide supporting empirical evidence in Section 3, calibrate the model in Section 4, and conduct counterfactual analyses in Section 5.

3. Cross-Country Supporting Evidence

This section presents cross-country correlations between fertility, structural change, and social norms. The analysis is motivated by Ngai and Petrongolo (2017) who argue that structural changes, in particular the service sector expansion, are gender-biased. We document the following two empirical regularities that are in line with the two predictions in Section 2.4:

Fact 1: Economies experiencing faster structural change, i.e., expansions of the service sector or declines in the agriculture sector, exhibit more rapid fertility declines.

Fact 2: The correlation between structural change and fertility is stronger in societies with tighter or less adaptive social norms.

3.1 Variable Construction

We combine different sources of data. Data on total fertility rate is obtained from the United Nations Population Division. Sectoral employment data are from the Groningen Growth and Development Centre (GGDC) 10-Sector Database (Timmer et al., 2015). This database offers long-term, internationally comparable data on value added and employment across ten sectors for more than 40 economies, including both developing and developed ones.[8] Data on gross national product (GDP) is collected from the Penn World Table 10.01.

To measure social norm tightness/pressure, we follow the literature on sociology and cultural studies. In particular, we obtain the tightness/looseness index from Uz (2015), where the author measures the tightness of social norms using the dispersion of opinions. The motivation of this measure is that in a tight culture, people's values, norms, and behavior are similar to each other because deviations are sanctioned. The merged sample includes 22 countries from all levels of economic development, with observations from the 1950s to 2010.

To measure social norm adaptability/stubbornness, we utilize data from the 2002 and 2012 International Social Survey Programme (ISSP) Family and Changing Gender Roles modules. Specifically, we analyzed responses to the statement, "A man's job is to

[8] We follow Herrendorf et al. (2014) in the sector assignments. In particular, Agriculture corresponds to the sum of International Standard Industrial Classification (ISIC) sections A–B, Manufacturing corresponds to the sum of ISIC sections C, D, F, and Services correspond to the sum of ISIC sections E, G–P.

earn money; a woman's job is to look after the home and family." Responses are coded as 1 if disagree, -1 if agree, and 0 if neutral. Two measures were created: "Norm Change: Total" captures the difference in average scores between the 1980 and 1920 birth cohorts, while "Norm Change Recent" reflects the 10-year inter-cohort change in gender attitudes. For instance, the change between 1960-1969 is determined by comparing the 1940-1949 and 1930-1939 birth cohorts.

To construct the speed of fertility change for country i, we run the regression

$$\text{tfr}_{i,\text{year}} = \alpha_i^{\text{tfr}} + \text{speed\_tfr}_i \times \text{year} + u_i \tag{13}$$

where speed_tfr_i measures the average annual change of the total fertility rate for country i during the sample period.

Likewise, we measure the speed of structural change for country i using

$$\text{service share}_{i,\text{year}} = \alpha_i^{\text{ser}} + \text{speed\_ser}_i \times \text{year} + v_i \tag{14}$$

where service share measures the fraction of service employment, and

$$\text{agriculture share}_{i,\text{year}} = \alpha_i^{\text{agr}} + \text{speed\_agr}_i \times \text{year} + v_i \tag{15}$$

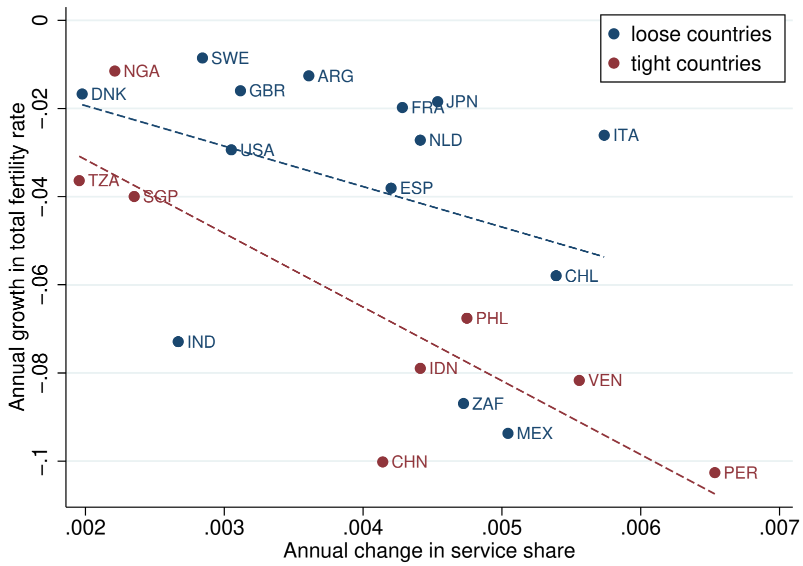
where agriculture share measures the fraction of agricultural employment. Therefore, speed_ser_i and speed_agr_i measures the average annual change of sectoral employment for country i during the sample period.

3.2 Structural Change and Fertility Decline

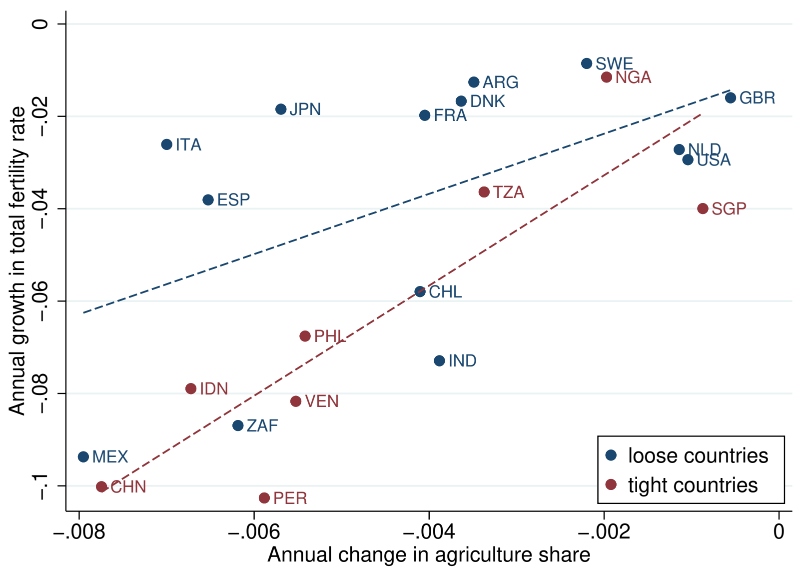
Figure 2a plots the correlation between speed_tfr_i and speed_ser_i. As can be seen, countries that have experienced faster expansion of the service sector, e.g., Venezuela and Peru, also witnessed more rapid fertility declines. Similarly, Figure 2b indicates that countries that have experienced faster shrinkage of the agriculture sector, e.g., China, and Mexico, also experienced more rapid fertility declines. The latter finding complements Ager et al. (2020) who document the relationship between agriculture decline and fertility transition in the American South.

Figure 2: Structural Change and Fertility Decline

(a) Service Expansion and Fertility Decline



(b) Agriculture Shrinkage and Fertility Decline



Notes: “speed_tfr” is defined as the average annual change in total fertility rate for each country in the sample period, using data from the United Nations Population Division. “speed_ser” and “speed_agr” are the average annual change in the fraction of employment in service and agriculture sectors respectively, using data from the Groningen Growth and Development Centre (GGDC).

3.3 The Role of Social Norm Tightness

To examine the relationship between structural change and fertility decline more systematically, we run a set of OLS regressions where we regress speed_tfr$_i$ on speed_SC$_i$ where SC (structural change) takes the value of speed_ser$_i$ or speed_agr$_i$, after controlling for the growth rate of GDP per capita speed_gdp$_i$. We also interact the speed of structural change with a dummy variable "tight", which takes the value of 1 if the country's opinion dispersion is below the median of the full sample in Uz (2015).

The regression results, presented in Table 1, reveal key insights. Column (1) shows that a 1-percentage-point increase in the speed of service sector expansion per year corresponds to an annual fertility decline of 0.13, a statistically and economically significant correlation. This relationship holds after controlling for GDP per capita growth in Column (2). In Column (3), we observe that the link between service sector expansion and fertility decline is predominantly driven by countries with tight social norms. Specifically, including countries with above-median cultural tightness increases the coefficient by more than 50%. Column (4) confirms that this heterogeneity persists even after adjusting for GDP per capita growth. Columns (5) through (8) replicate the analysis using the decline in agricultural employment as an alternative measure, yielding findings consistent with those for service sector expansion.

3.4 The Role of Social Norm Adaptability

We run panel data regressions to examine the relationships between fertility change, structural change speed, and social norm adaptability. Different from the cross-sectional specification, fertility and structural change speed are calculated as 10-year changes, based on the previously described datasets. GDP per capita level and growth were included as control variables.

The panel regression results, summarized in Table 2, complement our cross-sectional observations. We find that accelerated growth in service sector employment and a corresponding reduction in agricultural employment are significantly correlated with faster fertility declines, even after accounting for country fixed effects and year trends. Notably, a one-percentage-point increase in service sector share over a 10-year period is associated with a 0.07 decrease in the total fertility rate over the same time frame. Additionally, the interaction term between structural change speed and gender norm

Table 1: Regression Results: Cross-Sectional Data

	Dependent Variable: Fertility Change							
	Service				Agriculture			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
coef_ser	-13.40***	-13.58***	-9.54**	-9.66**				
	(4.64)	(4.74)	(4.30)	(4.44)				
tight=1 × coef_ser			-6.90**	-6.82**				
			(2.56)	(2.65)				
coef_agr					9.06***	9.18***	6.62***	6.74**
					(2.37)	(2.42)	(2.30)	(2.36)
tight=1 × coef_agr							5.10**	5.01**
							(1.98)	(2.03)
coef_gdp		0.19		0.09		0.21		0.14
		(0.39)		(0.35)		(0.35)		(0.32)
Observations	22	22	22	22	22	22	22	22
R-squared	0.294	0.303	0.489	0.491	0.422	0.433	0.572	0.576

Notes: "speed_tfr" is defined as the average annual change in total fertility rate for each country in the sample period, using data from the United Nations Population Division. "speed_SC" where SC $\in \{ser, agr\}$ are the average annual change in the fraction of employment in service and agriculture sectors respectively, using data from the Groningen Growth and Development Centre (GGDC). "speed_gdp" is calculated as the annual change in GDP per capita in the sample period, using data from the Penn World Table. "tight" is a dummy variable that takes the value of 1 if the tightness index is above the sample median, reflecting stronger constraints imposed by social norms on individual behavior. The index is developed by Uz (2015).

changes exhibits a positive sign for service sector growth, suggesting that the relationship between structural change and fertility is mitigated in societies with more adaptable social norms. The interaction term for agricultural share decline is negative, as the share of agriculture in the aggregate economy declines with economic development.

To summarize, we find that countries that have experienced faster structural change, i.e., service expansion or agriculture decline, witnessed a more rapid decline in fertility. Furthermore, this correlation is robust to controlling for the growth in GDP per capita, and is stronger in societies with tighter or less adaptive social norms. These findings are

Table 2: Regression Results: Panel Data

	Dependent Variable: Fertility Change							
	Service				Agriculture			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Speed_SC	-6.64***	-7.32***	-10.40***	-7.23***	7.66***	8.91***	9.58***	9.61**
	(0.70)	(0.74)	(1.57)	(0.91)	(0.53)	(0.62)	(0.92)	(0.63)
Speed_SC×Norm Change Total			5.35**				-1.94	
			(2.40)				(1.98)	
Speed_SC×Norm Change Recent				0.59				-0.49
				(0.38)				(0.31)
Norm Change Recent				0.59				-19.42***
				(0.38)				(4.09)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FEs	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Trend	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	785	785	785	785	785	785	785	785
R-squared	0.26	0.38	0.39	0.39	0.35	0.45	0.45	0.47

Notes: Fertility change is defined as the 10-year change in total fertility rate for each country in the sample period, using data from the United Nations Population Division. Structural change speed ("Speed_SC") is measured as the 10-year change in the service and agricultural employment shares, respectively, calculated using the Groningen Growth and Development Centre (GGDC) 10-sector database for years after 1950 and Mitchell (2007) for years before 1950. The variables "Norm Change Total" and "Norm Change Recent" are constructed from data obtained from the International Social Survey Programme: Family and Changing Gender Roles III (ISSP) 2002 and 2012. Both variables are based on responses to the question, "To what extent do you agree or disagree: A man's job is to earn money; a woman's job is to look after the home and family. " Responses are coded as 1 if disagree, -1 if agree, and 0 if neutral. "Norm Change Total" represents the difference in average scores between the 1980 and 1920 birth cohorts. "Norm Change Recent" measures the change in gender attitude of each birth cohort in a country compared to the cohort from 10 years prior. For example, if we examine the change between 1960-1969, then we are looking at the change in attitudes of the 1940-1949 birth cohort compared to the 1930-1939 birth cohort.. Control variables include GDP per capita level and growth.

in line with the predictions discussed in Section 2.4.

4. Calibration

In what follows, we apply the model to South Korea to analyze its fertility decline in recent decades. South Korea is a particularly interesting case because it has simultaneously experienced both exceptionally fast economic growth and structural transformation, and one of the most rapid declines in fertility since the 1960s. This section outlines the solution algorithm for the model, the calibration strategy, and presents the calibrated parameter values.

4.1 Model Solution

We first outline the solution algorithms for the steady state and the transition path.

Steady-State Solution The steady state is computed using an iterative method that converges to a social norm consistent with household decisions:

1. Guess an initial social norm η.
2. Given η, solve the childcare arrangement problem (3) subject to the childcare constraint (2), yielding optimal childcare functions $l^{♀}(n)$ and $l^{♂}(n)$.
3. Using $l^{♀}(n)$ and $l^{♂}(n)$, calculate each gender's outside option utility, $\overline{u}^g(n)$, for $g \in \{♀, ♂\}$.
4. With the outside options, solve the Nash bargaining problem for consumption $c^g(n)$ and utility $u^g(n)$, conditional on n.
5. Maximize each gender's utility, $n^g = \arg\max_n u^g(n)$, to determine desired fertility. The realized fertility is $n = \min\{n^{♀}, n^{♂}\}$, with associated childcare arrangements $l^{♀}(n)$ and $l^{♂}(n)$.
6. Compute the implied social norm, $\tilde{\eta} = l^{♀}(n)/l^{♂}(n)$. Update η and iterate until $|\eta - \tilde{\eta}| < \epsilon$, for a small tolerance $\epsilon > 0$.

Transition Path To compute the economy's dynamics, starting from any initial state (not necessarily the steady state), we use forward iteration:

1. Given the current social norm η_t, solve the static household optimization problem to obtain n_t, $l^{♀}(n_t)$, and $l^{♂}(n_t)$.
2. Using the current demographic structure $\boldsymbol{\pi_t}$ and fertility rate n_t, update the population distribution to $\boldsymbol{\pi_{t+1}}$ via Equation (12).

3. With $\boldsymbol{\pi_{t+1}}$ and the history of childcare arrangements $\{l^{♀}_{t-j}, l^{♂}_{t-j}\}_{j=0}^{J-J_f}$, calculate tomorrow's social norm η_{t+1} using Equation (9) after solving the every older cohorts' minimization problem (11). Repeat from step 1 for the next period.

This iterative process captures the co-evolution of fertility, childcare norms, and population structure over time.

4.2 Identification Strategy

We calibrate the model to replicate key data moments from South Korea over the period 1976–2016. The parameters to be calibrated are:

$$\underbrace{J, J_f}_{\text{demographics}}, \quad \underbrace{\gamma, \rho, \psi, \lambda}_{\text{preferences}}, \quad \underbrace{\phi, \beta, \sigma, \alpha}_{\text{technologies}}.$$

Several parameters are directly sourced from the literature. The economy of scale from spousal cooperation, α, is set to 1.2, following Doepke and Kindermann (2019). The time cost per child, ϕ, is fixed at 0.15, based on de La Croix and Doepke (2003). The elasticity of substitution between childcare, σ, is set to be 2.5 following Knowles (2013) and Yum (2023). We define each period as 5 years, setting $J = 16$ (total lifespan of 80 years) and $J_f = 6$ (childbearing between 25 to 30) to align with South Korea's life expectancy of approximately 80 years and a mean childbirth age of 25–30 during the study period.

The remaining parameters are calibrated to match the transition path of the South Korean economy from 1976 to 2016. In particular, we choose the exogenous time series for gender-specific wages, $w_t^{♀}$ and $w_t^{♂}$ to match the GDP per capita and gender wage gap statistics collected from the OECD database. Then, we simulate the path of the economy and ask the model to fit the observed trajectories of fertility, gender gaps in childcare responsibilities, and the share of within-cohort effects in social norm changes.[9]

We collect fertility data from the United Nations. Gender-specific childcare time is

[9] We also examined the ideal number of children by gender in the data (see Appendix B). We opted not to use these as target moments for two reasons. First, the survey question, 'What is the ideal number of children for a family to have?' may reflect respondents' views on societal norms or expectations for others rather than their personal fertility plans, i.e., social desirability bias (Bursztyn et al. (2025)). Second, the responses markedly diverge from actual fertility outcomes in both magnitude and trajectory. Nevertheless, we do observe a widening gender gap in the ideal number of children during periods of rapid structural transformation.

computed using the micro-level data from the Korea Time Use Survey (KTUS) following the strategy of Park (2021). Lastly, we use the micro-level data from the Korean General Social Survey (KGSS) to compute the share of within-cohort effects in driving social norm changes. To be more specific, for any variable Y, we first compute the change in average over time, i.e., the gap between $\overline{Y}_{\text{start time}}$ and $\overline{Y}_{\text{end time}}$. To compute within-cohort changes, we identify the cohorts that have more than N observations both at the start time and the end time. Then, we restrict the samples to the individuals from these cohorts, and compute how much Y changes over time in this sample, i.e., the gap between $\overline{Y}^{\text{subsample}}_{\text{start time}}$ and $\overline{Y}^{\text{subsample}}_{\text{end time}}$. The ratio between these two gaps measures the importance of within-cohort effects and is used as a targeted moment. In the KGSS data, we evaluate this ratio for three variables as Y:

1. SEXROLE1: Opinions on gender roles: It is more important for a wife to help her husband's career than to pursue her own career,
2. SEXROLE2: Opinions on gender roles: A husband's job is to earn money; a wife's job is to look after the home and family, and
3. HBBYWK08: Agree or disagree: Husband's job is to earn money; wife's job is to look after the home and family.

For all three cases, we find that the ratio is close to 0.8, implying an important role for within-cohort effects in driving the broader social norm change in South Korea in the study period.[10]

Although all parameters jointly influence the model's moments, certain moments provide stronger identification for specific parameters. Below, we outline the identification logic:

- The fertility weight, γ, is inferred from the initial fertility level in 1976, reflecting baseline preferences for children.
- The fertility curvature, ρ, governs the trade-off between consumption and fertility, identified by the fertility response to rising opportunity costs as wages increase for both genders.

[10] One concern is that these opinion measures are ordinal, not cardinal, complicating the mapping between the model's social norms and the data. We address this in multiple ways. First, we calibrate using the ratio of between-cohort changes to total changes, rather than opinion levels. Second, all three variables yield consistent results (around 0.8), reducing concerns about individual biases. Finally, sensitivity analysis with ratios from 0.6 to 1.0 shows the paper's main findings remain robust.

- The relative productivity in childcare, β, is determined by the initial gender gap in childcare time. Higher female productivity raises the initial gender gap by incentivizing greater specialization.
- The weight of individual's own experience in the formation of opinions, i.e., "stubbornness", ψ, is calibrated to match the share of between-cohort component in driving social norm changes. Smaller ψ leads to larger within-cohort effects.
- The social pressure parameter, λ, is calibrated to the persistence of gender gaps in childcare over time. A higher λ implies stronger resistance to change, as young couples face greater pressure from older cohorts' norms.

This strategy ensures the model captures both the levels and dynamics of fertility and childcare in South Korea over the calibration period.

4.3 Calibration Results

Figure 3 shows that the model closely replicates the observed trajectories of fertility and gender gaps in childcare responsibilities in South Korea from 1976 to 2016.

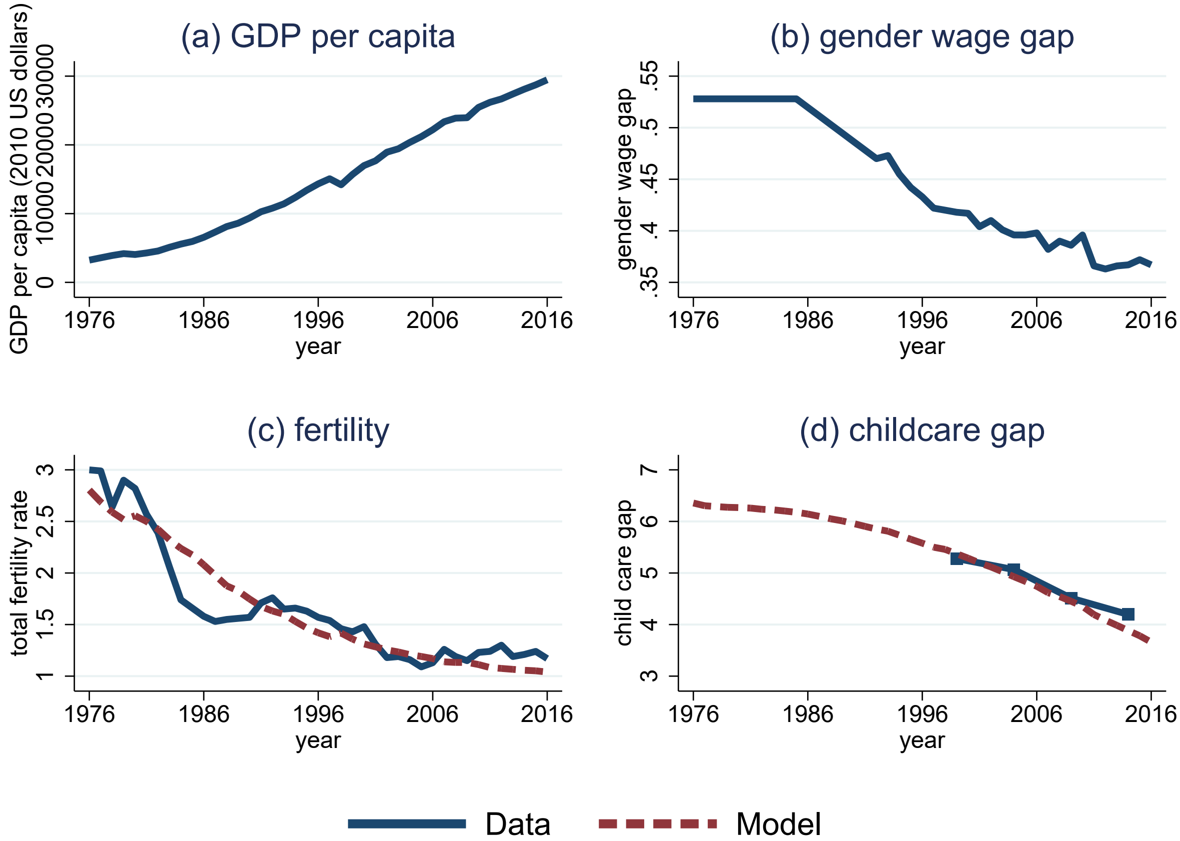
Table 3 presents the calibrated parameters alongside the corresponding data moments. Specifically, the calibrated childcare productivity of women, β, is close to 0.5, implying that women do not have a significant comparative advantage in childcare in the model. The social pressure parameter, $\lambda = 0.01$, yields utility losses for parents comparable to those in Myong et al. (2021) and Kim et al. (2024).

To illustrate these parameters' implications and the model's mechanisms, Table 4 compares childcare gaps and fertility across three scenarios.

The first scenario "old tech. & old norm" depicts a steady-state economy in 1976 with a large gender wage gap ($w^{♀}/w^{♂} = 0.47$). Here, women shoulder 6.36 times more childcare than men, consistent with the prevailing social norm ($\eta = 6.36$). Young couples' choices align with this norm, resulting in a fertility rate of $n = 2.8$.

The second scenario "new tech. & new norm" represents a new steady state with a narrower wage gap ($w^{♀}/w^{♂} = 0.63$). To highlight the mechanism, we let this to be achieved by raising female wage $w^{♀}$ from 0.47 to 0.63, while holding male wage $w^{♂}$ unchanged. Reduced wage disparity lowers the opportunity cost of men's childcare time, shifting the optimal allocation to a more equitable $l^{♀}/l^{♂} = 3.02$, which matches the updated norm ($\eta = 3.02$). Fertility dips slightly to $n = 2.65$, driven by the substitution effect

Figure 3: Calibration and Model Fit



Notes: Solid lines represent data. Dashed lines represent model-simulated values.

between consumption and children, as parameterized by ρ.

The third scenario "new tech. & old norm" explores a transitional case where technology narrows the wage gap to $w^{♀}/w^{♂} = 0.63$, but the social norm remains fixed at $\eta = 6.36$. Young couples, constrained by this norm, settle on a childcare division of $l^{♀}/l^{♂} = 6.12$, balancing monetary costs and utility losses. The unequal allocation worsens women's outside option, leading them to favor fewer children. With fertility determined by mutual agreement, the result is $n = 2.52$—notably lower than the long-run equilibrium in the second scenario. In other words, persistent social norm may amplify the direct effects of gender-biased technological changes by almost 80%.

Table 3: Calibrated Parameters

	Parameter	Value	Data moment	Source	Model fit
γ	Fertility weight	3.0	$n_{1976} = 3.0$	United Nations	3.0
σ	Childcare substitutability	2.5	Knowles (2013) and Yum (2023)		
β	Childcare productivity	0.5	$\eta_{1999} = 5.25$	Park (2021)	5.25
ρ	Fertility curvature	2.4	$n_{1976} \sim n_{2016}$	United Nations	See Figure 3
ψ	Stubbornness	4.5	Within-cohort effects	KGSS	80%
λ	Social pressure	0.01	$\eta_{1999} \sim \eta_{2014}$	Park (2021)	See Figure 3
α	Economies of scale	1.2	Doepke and Kindermann (2019)		
ϕ	Time costs per child	0.15	de La Croix and Doepke (2003)		
J	Total number of periods	16	80 years	World Health Organization	
J_f	The fertile period	6	25 to 30 yo	Statista	

Notes: This table presents the calibration results where the parameters are chosen to match the transition path of South Korea from 1976 to 2016.

Table 4: Three Different Cases

	Old tech. & old norm	New tech. & new norm	New tech. & old norm
$w^{\female}/w^{\male}$	0.47	0.63	0.63
η	6.36	3.02	6.36
$l^{\female}/l^{\male}$	6.36	3.02	6.12
n	2.80	2.65	2.52
Δn	-	-0.15	-0.28

Notes: $w^{\female}/w^{\male}$: gender wage ratio; η: social norm for mother's relative childcare time; $l^{\female}/l^{\male}$: actual relative childcare time chosen by the couple; n: fertility rate (children per couple). Δn is the change in fertility relative to the "old tech. & old norm" benchmark. "Old" corresponds to 1976 calibrated values; "new" corresponds to the long-run steady state after technological change. In the "new tech." scenarios the female wage rises while the male wage is held fixed.

5. Counterfactual Analysis

This section presents counterfactual experiments to examine how technological change, social norms, and policy interventions influence fertility, childcare allocation, and the evolution of social norms. To isolate the effects of technological progress operating through the gender wage gap, we hold men's wages $w_t^{♂}$ constant and allow variation only in women's wages $w_t^{♀}$. We study five scenarios, with results shown in Figures 4, 5, 6, 7, and 8. Each figure plots the trajectories of (a) the gender wage gap, (b) fertility, (c) the childcare gap, and (d) the social norm.

5.1 Speed of Structural Change

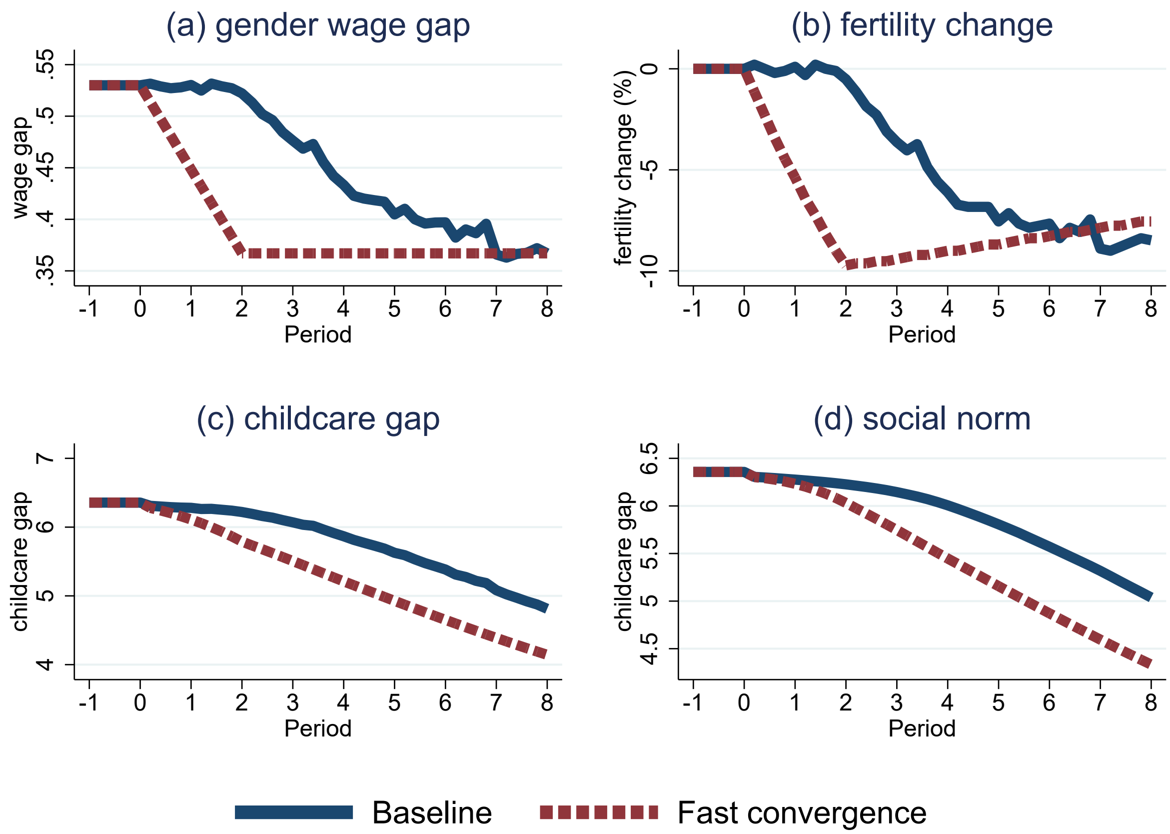
We first investigate how the pace of structural change affects economic dynamics by comparing a baseline path with a fast-convergence scenario. Starting from the 1976 steady state (gender wage gap of 53%, i.e., $w_t^{♀}/w_t^{♂} = 0.47$), we reduce the gap to 37% ($w_t^{♀}/w_t^{♂} = 0.63$), the level observed in South Korea by 2016, while keeping male wages $w_t^{♂}$ constant. In the baseline, this convergence unfolds over 40 years (1976–2016); in the fast scenario, it occurs in just 10 years. Results are shown in Figure 4.

In the fast-convergence case, the gender wage gap (panel a) falls sharply, outpacing the adjustment of the social norm (panel d), which remains close to its initial value of 6.36 for several years because of slow cohort replacement and limited re-evaluation by older cohorts. This mismatch triggers a rapid fertility decline (panel b) of 10% within a decade, as women face higher opportunity costs yet operate under a norm that still enforces highly unequal childcare (panel c, $l_t^{♀}/l_t^{♂} \approx 6.0$). Over time, as new cohorts adopt more equitable allocations, the childcare gap and social norm converge toward 4, and fertility gradually recovers.

The childcare gap ($l_t^{♀}/l_t^{♂}$) and social norm (η_t) follow nearly identical paths in both scenarios and closely match the calibrated baseline. This similarity arises from strong social pressure ($\lambda = 0.01$), which heavily penalizes deviations from tradition, and limited re-evaluation by older cohorts ($\psi = 4.5$). Even as women's wages rise, young couples adjust childcare only incrementally, constrained by the utility cost of defying a norm shaped by older generations whose expressed opinions resemble their own past behavior. The main difference therefore appears in the speed of the fertility decline, highlighting fertility's acute sensitivity to the timing of shifts in the relative opportunity cost of children.

These results (Figure 4) reveal a non-monotonic fertility response: rapid technological progress amplifies short-run fertility declines by intensifying the conflict between economic incentives and rigid social constraints, whereas gradual norm adjustment

Figure 4: The Role of the Speed of Structural Change



Notes: Solid lines represent the baseline case. Dashed lines represent counterfactual results.

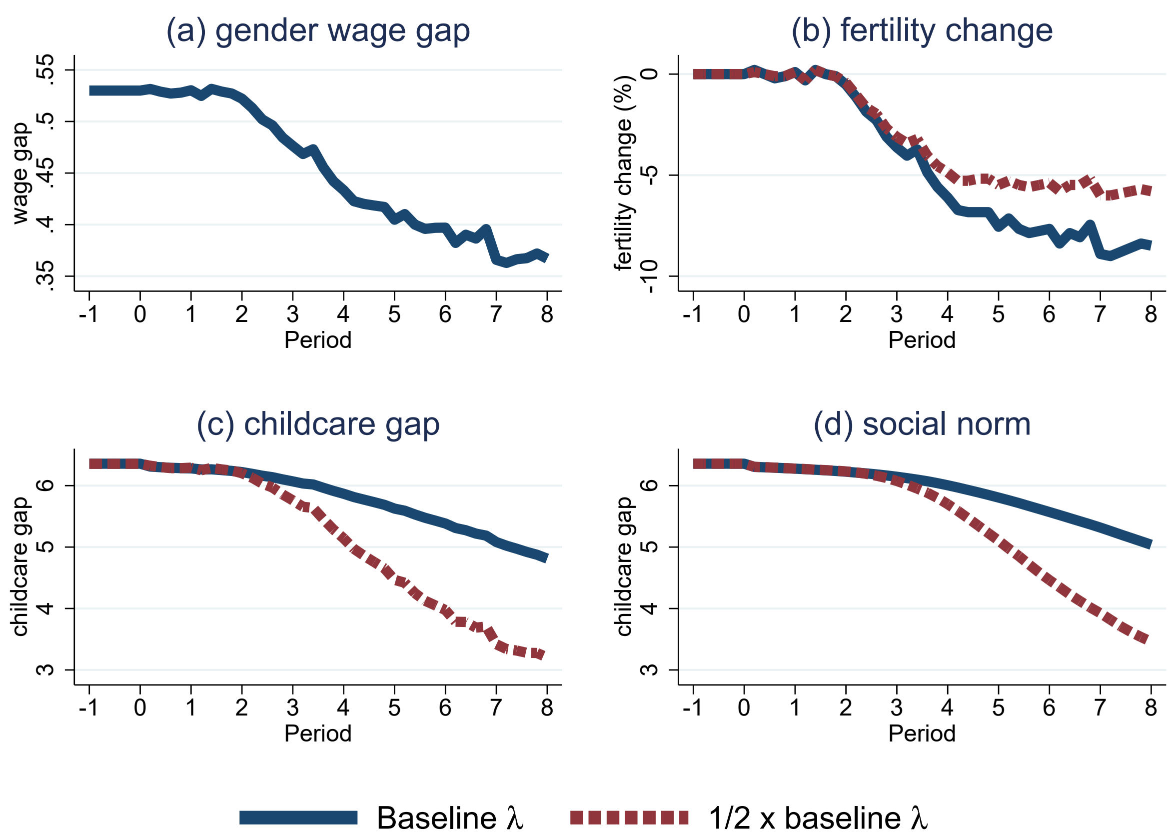
eventually permits fertility recovery.

5.2 The Role of Social Pressure

We next evaluate how the intensity of social pressure shapes the response to structural change. We simulate the baseline 40-year wage-gap convergence (53% to 37%) under the calibrated social-pressure parameter $\lambda = 0.01$ and under a weaker value $\lambda = 0.005$. Results appear in Figure 5.

With weaker social pressure, the decline in the gender wage gap (panel a) triggers a rapid narrowing of the childcare gap (panel c), since couples face lower utility costs for egalitarian allocations. The social norm (panel d) adjusts quickly, falling to 5 by 2001. Consequently, the fertility decline is considerably muted, amounting to only 5% by 2016 compared with 8% in the baseline.

Figure 5: The Role of Social Pressure



Notes: Solid lines represent the baseline case. Dashed lines represent counterfactual results.

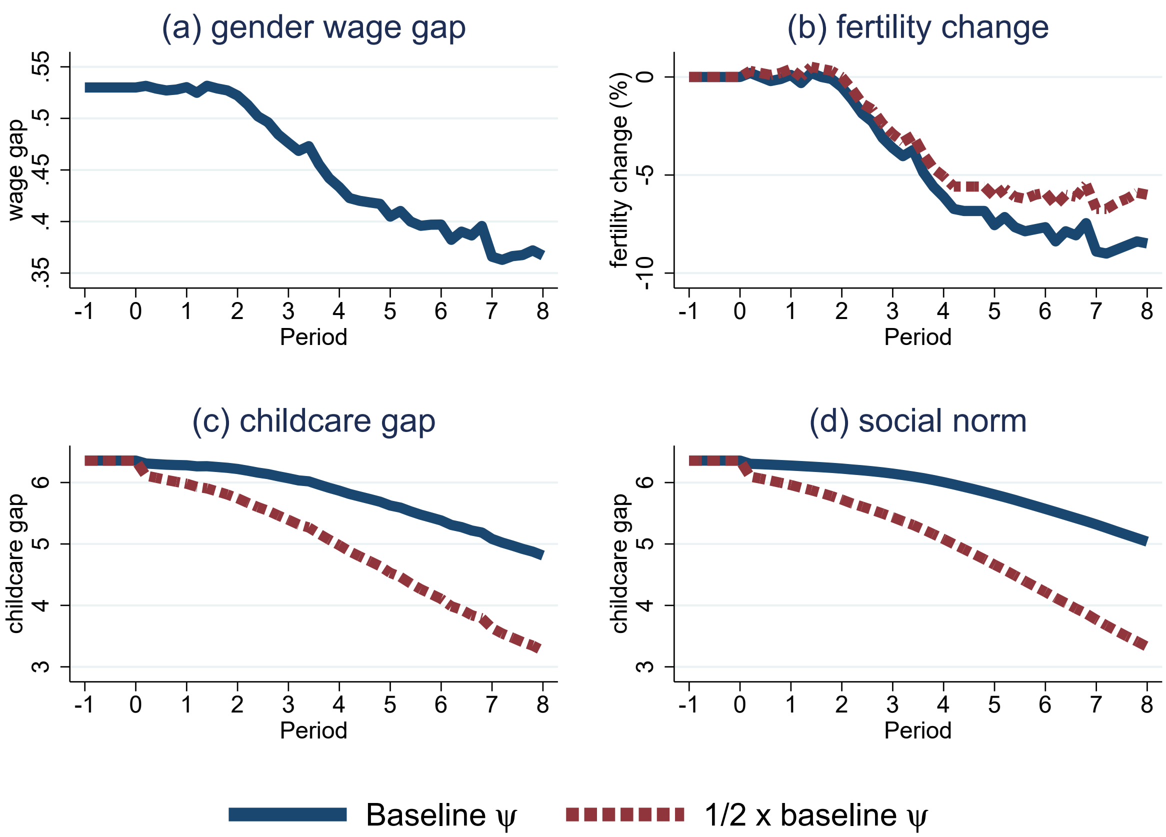
Figure 5 thus identifies social pressure as a key amplifier of technological shocks. Lower λ eases the tension between rising female wages and lagging norms, yielding a smoother transition and higher fertility.

5.3 The Role of Older Cohorts' Re-evaluation

We then examine how the willingness of older cohorts to re-evaluate their expressed opinions affects the transition. We repeat the baseline wage-gap convergence under the calibrated "stubbornness" parameter $\psi = 4.5$ and under a lower value $\psi = 2.25$. Results are displayed in Figure 6.

Lower stubbornness allows older cohorts to voice more egalitarian opinions, reducing the utility penalty imposed on young couples. This accelerates the decline in the childcare gap (panel c) and speeds convergence of the social norm (panel d). Fertility therefore falls by only 6% by 2016, versus 8% in the baseline.

Figure 6: The Role of Older Cohorts' Re-evaluation



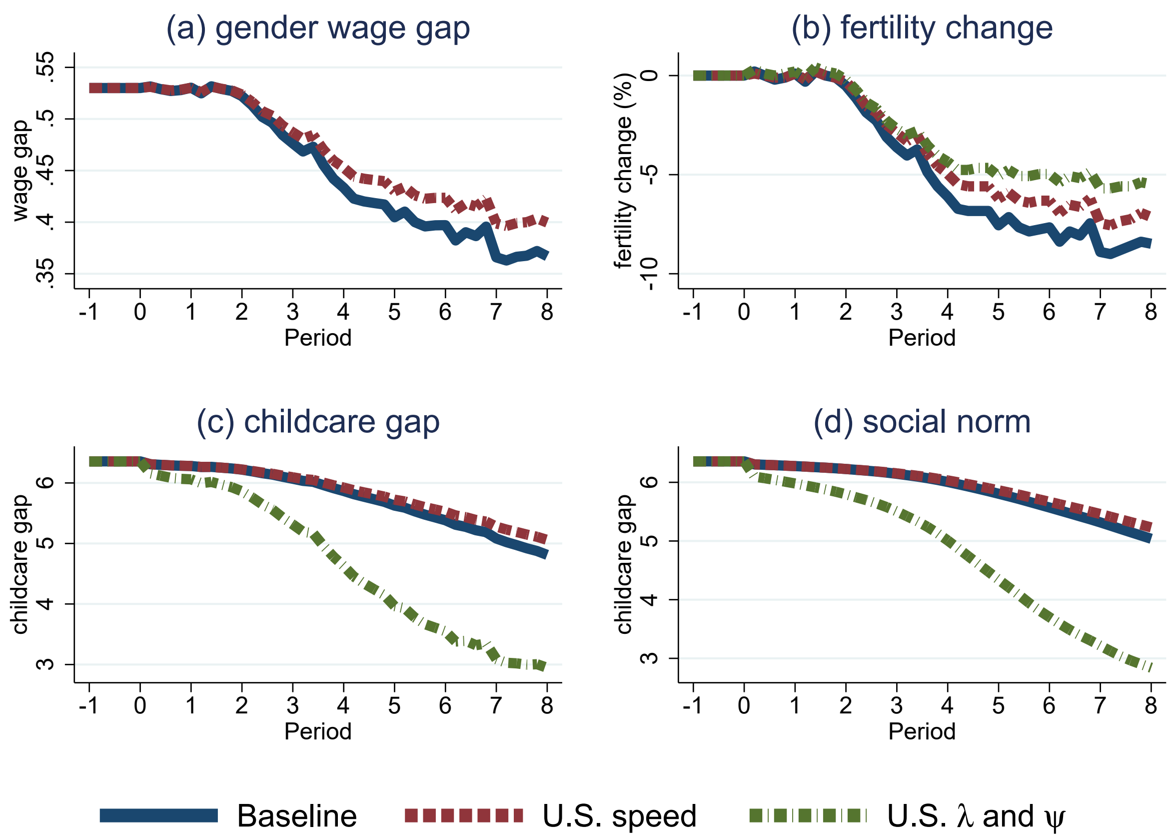
Notes: Solid lines represent the baseline case. Dashed lines represent counterfactual results.

Figure 6 confirms that limited re-evaluation by older cohorts acts as another important friction amplifying technological shocks. Lower ψ mitigates the conflict between economic opportunities and inherited norms, supporting higher fertility.

5.4 U.S. Parameters

We also explore a hypothetical Korean fertility transition using parameters calibrated to the U.S. experience from 1965 to 2015 (details in Section A). Two counterfactuals are compared with the Korean baseline: (i) the slower U.S. speed of gender-wage-gap convergence, and (ii) U.S.-like social-pressure and stubbornness parameters ($\lambda = 0.005$, $\psi = 2.0$). Results appear in Figure 7.

Figure 7: Counterfactual Korean Transition with U.S. Parameters



Notes: Solid lines represent the Korean baseline. Dashed lines represent counterfactual results.

Slower wage-gap convergence delays and smooths the transitions in fertility, childcare, and norms, consistent with Section 5.1. Lower social pressure and stubbornness produce a markedly milder fertility decline and accelerate adjustment of the childcare gap (panel c) and social norm (panel d), in line with Sections 5.2 and 5.3.

5.5 Gender-Specific Childcare Subsidies

Many developed economies have introduced childcare subsidies and paid parental leave to promote workplace gender equality and support fertility (e.g., Duvander and Andersson 2006; Lalive and Zweimüller 2009; Farré and González 2019). Our framework is ideally suited to study the short- and long-run effects of such policies.

We analyze a lump-sum-tax-financed policy that reimburses part of the opportunity cost of childcare, comparing subsidies targeted at mothers versus fathers. The objective function in the childcare allocation problem (Equation (3)) becomes, for female subsidies:

$$\min_{l^{♀},l^{♂}} (1-\tau^{♀})w_t^{♀}l^{♀} + w_t^{♂}l^{♂} + \lambda w_t^{♂}\left(\frac{l^{♀}}{l^{♂}} - \eta_t\right)^2, \tag{16}$$

and for male subsidies:

$$\min_{l^{♀},l^{♂}} w_t^{♀}l^{♀} + (1-\tau^{♂})w_t^{♂}l^{♂} + \lambda w_t^{♂}\left(\frac{l^{♀}}{l^{♂}} - \eta_t\right)^2. \tag{17}$$

We calibrate $\tau^{♀} = 10\%$ and $\tau^{♂} = \tau^{♀} \cdot \frac{w^{♀}l^{♀}}{w^{♂}l^{♂}}$ so that both policies have identical fiscal cost in the initial steady state before behavioral responses. Figure 8 compares the two policies with the no-policy baseline.

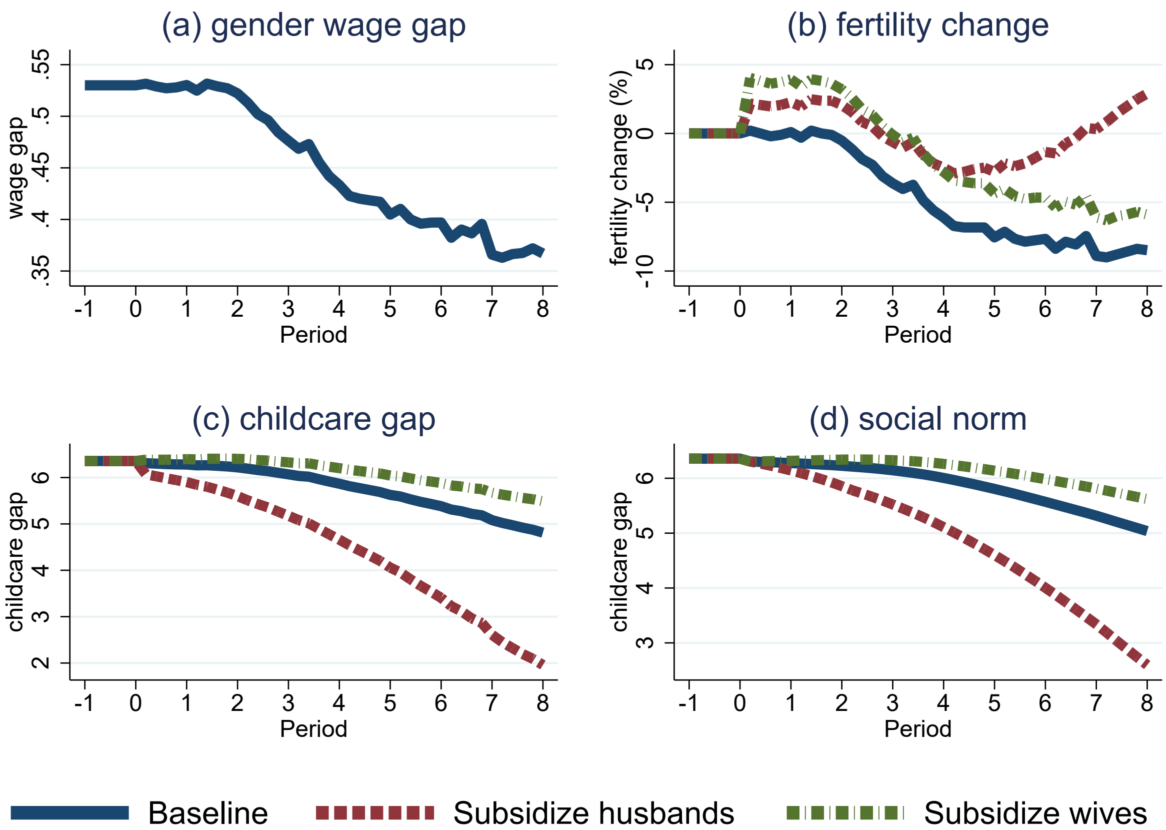
Both policies raise fertility relative to the baseline (panel b), but through different channels and with divergent long-run consequences. Subsidizing female childcare generates a large immediate fertility boost sufficient to offset the early phase of gender-biased technological change. However, it widens the childcare gap and reinforces traditional norms (panels c and d), so fertility gains erode as the wage gap continues to narrow.

Subsidizing male childcare produces a smaller short-run fertility increase, limited by initial norm resistance. Yet it narrows the childcare gap, hastens the shift toward egalitarian norms, and thereby delivers larger fertility gains after approximately 20 years (panel b). Two mechanisms drive this reversal: (i) the effective shadow price of children falls progressively under male subsidies as fathers take on more childcare, whereas it rises again under female subsidies; (ii) faster norm equalization reduces the long-run shadow price of children when women's relative wages are higher (see Figure 1).

Figure 8 illustrates the importance of embedding fertility policy analysis in a setting with evolving social norms. Although female-targeted subsidies appear more effective in the short run, male-targeted subsidies ultimately dominate because they accelerate the transition to a low-fertility, high-norm-cost steady state.

These predictions align with recent evidence. Lee and Lee (2025) study a South Ko-

Figure 8: Gender-Specific Childcare Subsidies



Notes: Solid lines represent the no-policy baseline. Dashed lines represent counterfactual results.

rean conglomerate's introduction of fully paid paternal leave and document sharply higher male take-up, a breakdown of traditional childcare norms, and substantial fertility increases—especially among high-earning wives—consistent with our mechanism.[11] Similarly, Kleven et al. (2025) show that Denmark's earmarked paternity leave shifted gender norms and childcare practices, and Albrecht et al. (2024) model endogenous peer pressure in leave take-up, underscoring the dynamic feedback our framework captures.

[11] Kim and Yum (2025) also find, in a life-cycle model without endogenous norms, that incentivizing male leave yields larger fertility effects.

6. Conclusion

Fertility rates are declining globally, posing significant economic and demographic challenges. This paper develops a quantitative model of fertility bargaining, where equilibrium outcomes emerge from the interplay between gender-biased technological progress and endogenous social norms shaped by past cohorts' opinions, leading to both within- and between-cohort forces of social norm changes. Consistent with the model's predictions, we document that fertility falls more rapidly in economies undergoing rapid structural change, with the decline amplified in societies with tighter or less adaptive social norms.

Calibrated to South Korea's experience from 1976 to 2016, our model closely matches the observed trajectories of fertility and childcare gaps, capturing the tension between wage convergence and norm inertia. Counterfactual analyses highlight the pivotal role of adjustment speed: rapid technological change outpaces norm evolution, driving sharper fertility drops, while weaker social pressure, more re-evaluation by older cohorts, or norm-shifting policies (e.g., subsidies to male childcare) mitigate these declines and expedite fertility recovery.

While the model is calibrated to match South Korea's growth experience, we contend that its lessons offer valuable guidance for many developing countries navigating, or on the cusp of, rapid structural transformation. Reconciling gender-biased technological progress with entrenched social norms is essential to tackling the "getting old before getting rich" dilemma these countries face.

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A. Calibration for the United States

In this section, we calibrate the model to match the transition path of fertility and the gender childcare gap observed in the United States from 1965 to 2015. The identification strategy follows from Section 4.2.

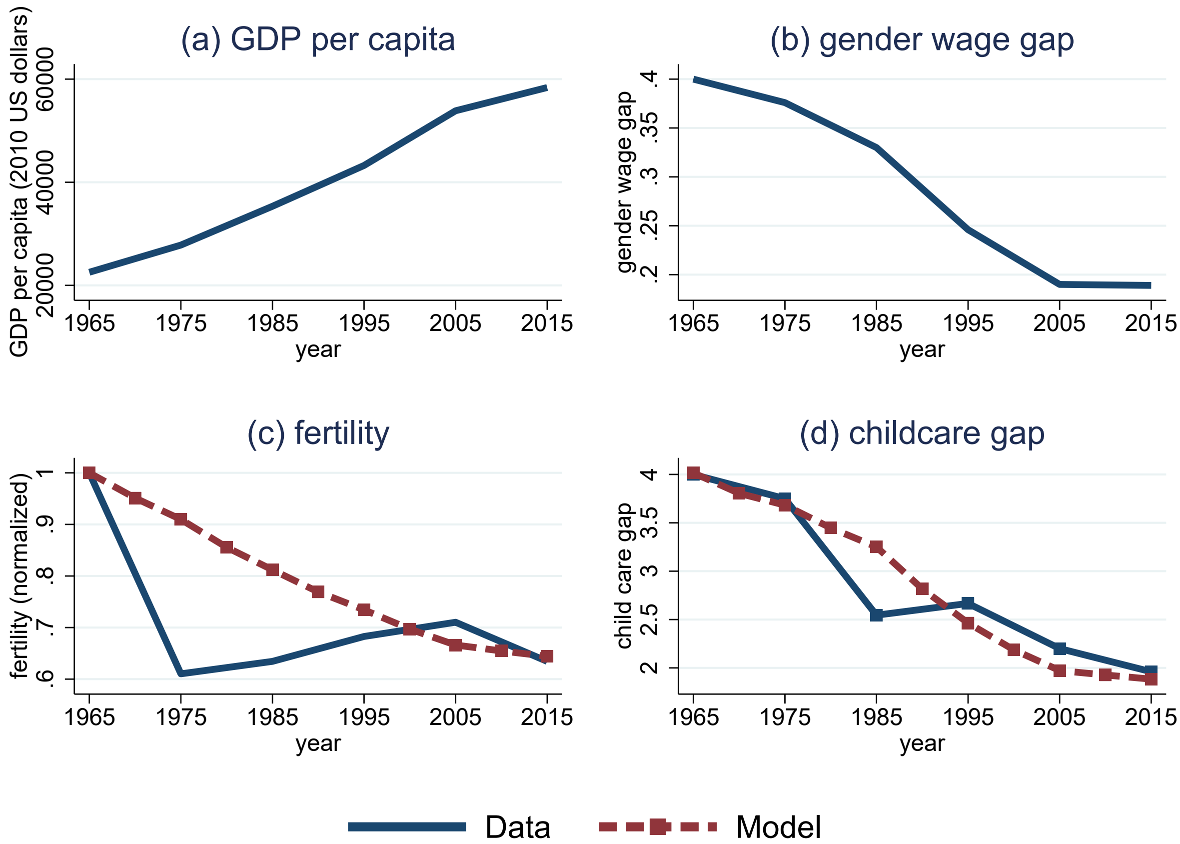
We collect fertility data from the United Nations. Gender-specific childcare time between 1965 and 1995 is collected from Egerton et al. (2005) who utilizes the micro-level data from the American Heritage Time Use Study (AHTUS). After 2003, gender-specific childcare time is calculated using the micro-level data from the American Time Use Survey (ATUS) following the strategy of Milkie et al. (2025). Lastly, we use the micro-level data from the General Social Survey (GSS) to compute the share of within-cohort effects in driving social norm changes. We apply the same method as described in Section 4.2 to the variable "*fefam*: It is much better for everyone involved if the man is the achiever outside the home and the woman takes care of the home and family." We find that the ratio of within-cohort change divided by total change is close to 0.3 over the course of 50 years, similar to the decomposition result by Brooks and Bolzendahl (2004).

Table 5: Calibrated Parameters - USA

	Parameter	Value	Data moment	Source	Model fit
γ	Fertility weight	2.27	$n_{1965} = 2.90$	United Nations	2.90
σ	Childcare substitutability	2.5	Knowles (2013) and Yum (2023)		
β	Childcare productivity	0.51	$\eta_{1965} = 4.0$	Egerton et al. (2005)	4.0
ρ	Fertility curvature	2.4	$n_{1965} \sim n_{2015}$	United Nations	See Figure 9
ψ	Stubbornness	2.0	Within-cohort effects	GSS	30%
λ	Social pressure	0.005	$\eta_{1965} \sim \eta_{2015}$	Egerton et al. (2005)	See Figure 9
α	Economies of scale	1.2	Doepke and Kindermann (2019)		
ϕ	Time costs per child	0.15	de La Croix and Doepke (2003)		
J	Total number of periods	16	80 years	World Health Organization	
J_f	The fertile period	6	25 to 30 yo	Statista	

Figure 9 plots the time path of GDP per capita and the gender wage gap as inputs into the transition path calibration. Panels (c) and (d) of 9 present the data and model-

Figure 9: Calibration and Model Fit - USA



generated path of fertility and the childcare gap. As can be seen, while the model does not fully account for the decline and recovery of fertility after the baby boom, it fits the overall decline in fertility and the path of the childcare gap in the study period well.[12]

Table 5 displays the calibrated parameters for the U.S. economy. Except for the parameter γ which governs the level of fertility, parameters β and ρ are very similar to the results in South Korea (see Table 3). The parameters governing the social norm evolution, however, are quite different. The value for λ in the U.S. is 17% smaller than that in South Korea, indicating a smaller magnitude of social pressure. In addition, the value for ψ in the U.S. is 33% smaller than that in South Korea, implying a smaller degree of stubbornness and hence greater re-evaluation by older cohorts.

[12] See Greenwood et al. (2005) for an in-depth study of the U.S. baby boom. The rising immigration from sending countries with high fertility, e.g., Mexico, has also contributed to the fertility recovery since the 1970s (Jonsson and Rendall (2004)).

B. Ideal Number of Children by Gender

This section displays the average responses to the question, IDLCHDN: ”What is the ideal number of children for a family to have?” from the Korean General Social Survey (KGSS), broken down by survey year and respondent gender. The sample includes only respondents aged 24 to 34, a 10-year range centered on the median age of first birth.

Table 6: Ideal Number of Children

Survey Year	Male respondents	Female respondents
2006	2.41	2.38
2008	2.52	2.45
2012	2.64	2.41

Notes: This table displays the average responses to the question, IDLCHDN: ”What is the ideal number of children for a family to have?” from the Korean General Social Survey (KGSS), broken down by survey year and respondent gender. The sample includes only respondents aged 24 to 34, a 10-year range centered on the median age of first birth.